ASSIGNMENT 9A

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[1]:
```
# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

1. Load the Mall Customers Dataset

[ ]:
```
# Load the Mall Customers dataset from a direct CSV link
# If one source is unavailable, the code tries a backup source.
urls = [
    "https://raw.githubusercontent.com/sharmaroshan/Clustering-of-Mall-Customers/
  ↪master/Mall_Customers.csv",
    "https://raw.githubusercontent.com/tirthajyoti/Machine-Learning-with-Python/
  ↪master/Datasets/Mall_Customers.csv",
]

df = None
for url in urls:
    try:
        df = pd.read_csv(url)
        print("Dataset loaded successfully.")
        break
    except Exception as error:
        print("Could not load from this source:", url)
        print("Trying another source...")

if df is None:
    raise RuntimeError(
        "Could not download the dataset. Check your internet connection, "
        "or download Mall_Customers.csv and load it using pd.
  ↪read_csv('Mall_Customers.csv')."
    )

print("First five rows:")
display(df.head())
print("Dataset shape:", df.shape)
```

2. Select Features

[ ]:
```
# Use Annual Income and Spending Score for clustering
X = df[["Annual Income (k$)", "Spending Score (1-100)"]]

print("Features used for clustering:")
display(X.head())
```

3. Find the Optimal Number of Clusters Using the Elbow Method

[ ]:
```
inertia = []

for k in range(1, 11):
    model = KMeans(n_clusters=k, init="k-means++", random_state=42, n_init=10)
    model.fit(X)
    inertia.append(model.inertia_)

plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), inertia, marker="o", linestyle="--")
plt.title("Elbow Method")
plt.xlabel("Number of Clusters (k)")
plt.ylabel("Inertia (WSS)")
plt.xticks(range(1, 11))
plt.grid(True)
plt.tight_layout()
plt.show()
```

4. Apply K-Means with 5 Clusters

[ ]:
```
# The elbow graph for this dataset commonly suggests 5 clusters
optimal_k = 5

kmeans = KMeans(
    n_clusters=optimal_k, init="k-means++", random_state=42, n_init=10
)
df["Cluster"] = kmeans.fit_predict(X)

print(f"K-Means clustered the customers into {optimal_k} groups.")
display(df[["Annual Income (k$)", "Spending Score (1-100)", "Cluster"]].head())

# Graph of the 5 clusters found by K-Means
plt.figure(figsize=(8, 5))
scatter = plt.scatter(
    df["Annual Income (k$)"],
    df["Spending Score (1-100)"],
    c=df["Cluster"],
```

```
    cmap="viridis",
    alpha=0.8
)
plt.title("K-Means with 5 Clusters")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.colorbar(scatter, ticks=range(optimal_k), label="Cluster")
plt.grid(True)
plt.tight_layout()
plt.show()
```

5. Visualize the Customer Clusters

[ ]:
```
plt.figure(figsize=(8, 5))

for cluster in range(optimal_k):
    cluster_data = df[df["Cluster"] == cluster]
    plt.scatter(
        cluster_data["Annual Income (k$)"],
        cluster_data["Spending Score (1-100)"],
        label=f"Cluster {cluster + 1}",
        alpha=0.7
    )

# Plot the cluster centers
centers = kmeans.cluster_centers_
plt.scatter(
    centers[:, 0], centers[:, 1],
    s=200, marker="X", label="Centroids"
)

plt.title("K-Means Customer Clusters")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

6. Cluster Analysis

[ ]:
```
# Show number of customers and average values in each cluster
cluster_summary = df.groupby("Cluster").agg(
    Number_of_Customers=("Cluster", "size"),
    Average_Annual_Income=("Annual Income (k$)", "mean"),
    Average_Spending_Score=("Spending Score (1-100)", "mean")
```

```
).round(2)

display(cluster_summary)
```

Conclusion

The K-Means algorithm was applied to the Mall Customers dataset using annual income and spending score. The Elbow Method was used to inspect a suitable number of clusters, and the customers were grouped into five clusters. The scatter plot and cluster summary help compare the customer groups.